Calculus I Coursebook

Definitions, Properties, Examples, Exercises, and Spaced Repetition Across Lessons

Purpose. This coursebook is designed to teach introductory calculus while making spaced repetition visible in the teaching sequence. Each new topic includes:

- core definitions,
- key properties and rules,
- fully worked examples,
- exercise sets,
- and a **Spaced Repetition Block** that deliberately brings back earlier content.

How spacing is integrated

This file uses a simple instructional rhythm:

1. **First contact:** teach the new idea with support and worked examples.
2. **Short return:** revisit it later in the same lesson through a retrieval prompt.
3. **Next-lesson return:** begin the following lesson with 2–4 questions from earlier topics.
4. **Delayed cumulative return:** older ideas reappear inside new contexts, not only in isolated review.
5. **Mixed assessment:** exercise sets never remain purely blocked after the first lesson.

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Course Structure and Spacing Map

Lesson	New Topic	Planned Returns
1	Limits	Same-lesson retrieval, cumulative return in Lessons 2–6
2	Continuity	Warm-up on limits; continuity exercises use limit ideas directly
3	Derivative as a limit	Warm-up on limits and continuity; derivatives built from earlier material
4	Derivative rules and applications	Mixed derivative practice plus returns to limits and continuity
5	Integrals	Warm-up on derivatives; integrals compared with rates of change
6	Cumulative review and synthesis	All prior topics mixed in one structured revision set

Teaching Note

The key principle is not "repeat everything constantly." The key principle is: *bring back what future learning depends on.* In calculus, that means limits never disappear, continuity stays active, and derivatives keep returning even when the topic has moved to applications or integrals.

Lesson 1 Limits of Functions

Lesson Goals

By the end of this lesson, students should be able to:

- explain the intuitive meaning of a limit,
- compute limits from tables, graphs, and formulas,
- identify when left-hand and right-hand limits agree or disagree,
- use basic limit laws in simple situations.

Definitions

Definition 1.1. *Let f be a function defined near a. We say that*

$$\lim_{x \to a} f(x) = L$$

if the values of $f(x)$ get arbitrarily close to L whenever x gets sufficiently close to a (but not necessarily equal to a).

Definition 1.2. *The **left-hand limit** of f at a is*

$$\lim_{x \to a^-} f(x),$$

*and the **right-hand limit** of f at a is*

$$\lim_{x \to a^+} f(x).$$

The two-sided limit exists only if both one-sided limits exist and are equal.

Properties

Property 1.1 (Limit Laws)**.** *If $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$, then:*

$$\begin{aligned}
\lim_{x \to a}[f(x) + g(x)] &= L + M, \\
\lim_{x \to a}[f(x) - g(x)] &= L - M, \\
\lim_{x \to a}[f(x)g(x)] &= LM, \\
\lim_{x \to a} \frac{f(x)}{g(x)} &= \frac{L}{M} \quad \text{if } M \neq 0.
\end{aligned}$$

Property 1.2 (Polynomial and Rational Functions)**.** *If $p(x)$ is a polynomial, then*

$$\lim_{x \to a} p(x) = p(a).$$

If $r(x) = \frac{p(x)}{q(x)}$ is rational and $q(a) \neq 0$, then

$$\lim_{x \to a} r(x) = r(a).$$

Worked Examples

Example 1.1. *Evaluate*

$$\lim_{x\to 3}(2x+5).$$

Solution. *Since $2x+5$ is a polynomial expression, we substitute directly:*

$$\lim_{x\to 3}(2x+5) = 2(3)+5 = 11.$$

Example 1.2. *Evaluate*

$$\lim_{x\to 2}\frac{x^2-4}{x-2}.$$

Solution. *Direct substitution gives $\frac{0}{0}$, so we factor:*

$$x^2-4 = (x-2)(x+2).$$

For $x \neq 2$,

$$\frac{x^2-4}{x-2} = x+2.$$

Therefore,

$$\lim_{x\to 2}\frac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4.$$

Example 1.3. *Consider*

$$f(x) = \begin{cases} 1, & x<0, \\ 3, & x\geq 0. \end{cases}$$

Find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

Solution. *From the left, the function stays equal to 1, so*

$$\lim_{x\to 0^-} f(x) = 1.$$

From the right, the function stays equal to 3, so

$$\lim_{x\to 0^+} f(x) = 3.$$

Since these are not equal, the two-sided limit does not exist.

Exercise Set

A. Direct practice

1. Compute $\lim_{x\to 1}(4x-7)$.
2. Compute $\lim_{x\to -2}(x^2+3x+1)$.
3. Compute $\lim_{x\to 4}\frac{2x+1}{x-1}$.

B. Indeterminate forms

4. Compute $\lim_{x\to 5}\frac{x^2-25}{x-5}$.
5. Compute $\lim_{x\to -1}\frac{x^2+3x+2}{x+1}$.

C. One-sided reasoning

6. Define a piecewise function with different left-hand and right-hand limits at $x=2$ and state whether the two-sided limit exists.

7. Explain in words why a function may fail to have a limit at a point even when it is defined there.

Retrieval Prompt

Without looking back, write the difference between a limit and the actual function value at a point. Then give one example where they are different.

Spaced Repetition Block

Same lesson return.

- End the lesson with two quick questions: one direct-substitution limit and one factor-and-cancel limit.

Planned next return.

- Begin Lesson 2 with three limit questions: one polynomial, one rational, and one one-sided limit.

Why this matters.

- Limits are not a disposable prelude. They become the language of continuity and the definition of the derivative.

Lesson 2 Continuity

Lesson Goals

By the end of this lesson, students should be able to:

- state the definition of continuity at a point,
- test continuity using the three standard conditions,
- classify discontinuities in simple examples,
- explain how continuity depends on limits.

Definitions

Definition 2.1. *A function f is* ***continuous at*** $x = a$ *if:*

1. $f(a)$ *exists,*
2. $\lim_{x\to a} f(x)$ *exists,*
3. $\lim_{x\to a} f(x) = f(a)$.

Definition 2.2. *A function is* ***continuous on an interval*** *if it is continuous at every point of that interval.*

Properties

Property 2.1. *Polynomials are continuous everywhere on* $\mathbb{R}$.

Property 2.2. *Rational functions are continuous everywhere they are defined.*

Property 2.3. *If f and g are continuous at a, then $f+g$, $f-g$, and fg are continuous at a. Also, $\frac{f}{g}$ is continuous at a provided $g(a) \neq 0$.*

Worked Examples

Example 2.1. *Show that $f(x) = x^2 - 4x + 1$ is continuous at $x = 2$.*

Solution. *Since f is a polynomial, it is continuous everywhere. Explicitly,*

$$f(2) = 2^2 - 4(2) + 1 = -3,$$

and

$$\lim_{x\to 2}(x^2 - 4x + 1) = -3.$$

Thus $\lim_{x\to 2} f(x) = f(2)$, so f is continuous at 2.

Example 2.2. *Consider*

$$f(x) = \frac{x^2 - 9}{x - 3}.$$

Is f continuous at $x = 3$?

Solution. *The function is not defined at $x = 3$, so the first continuity condition fails immediately. Even though*

$$\frac{x^2 - 9}{x - 3} = x + 3 \quad \text{for } x \neq 3,$$

and

$$\lim_{x\to 3} f(x) = 6,$$

the function is not continuous at $x = 3$ because $f(3)$ does not exist.

Example 2.3. *Let*

$$f(x) = \begin{cases} x+1, & x < 1, \\ 5, & x = 1, \\ 2x, & x > 1. \end{cases}$$

Determine whether f is continuous at $x = 1$.

Solution.

$$\lim_{x\to 1^-} f(x) = 2, \qquad \lim_{x\to 1^+} f(x) = 2.$$

So the limit exists and equals 2. But $f(1) = 5$, hence

$$\lim_{x\to 1} f(x) \neq f(1).$$

Therefore, f is not continuous at $x = 1$.

Exercise Set

A. Check continuity

1. Is $f(x) = 3x^3 - 2x + 7$ continuous at $x = 0$?
2. Is $f(x) = \frac{x+1}{x^2-4}$ continuous at $x = 1$?
3. Is $f(x) = \frac{1}{x-2}$ continuous at $x = 2$?

B. Analyze piecewise functions

4. For

$$f(x) = \begin{cases} x^2, & x \leq 1, \\ 2x - 1, & x > 1, \end{cases}$$

determine whether f is continuous at $x = 1$.

5. Construct a piecewise function with a removable discontinuity at $x = 4$.

C. Explanation

6. Explain why continuity cannot be checked without using limits.
7. Describe the difference between a removable discontinuity and a jump discontinuity.

Retrieval Prompt

From memory, list the three conditions for continuity at a point.

Spaced Repetition Block

Warm-up from Lesson 1.

1. Compute $\lim_{x\to 2}(x^2 + 1)$.
2. Compute $\lim_{x\to 1} \frac{x^2-1}{x-1}$.
3. Give an example where the left-hand and right-hand limits are different.

Return strategy.

- In Lesson 3, begin with one continuity question and one limit question.
- When discussing the derivative definition, explicitly ask: why do limits still matter here?

Lesson 3 The Derivative as a Limit

Lesson Goals

By the end of this lesson, students should be able to:

- state the limit definition of the derivative,
- compute derivatives from first principles in simple cases,
- connect differentiability to continuity,
- explain why derivatives are built from limits rather than direct formulas alone.

Definitions

Definition 3.1. *The **derivative** of f at $x = a$ is*

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided the limit exists.

Definition 3.2. *The **derivative function** is*

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

when this limit exists for each x in the domain considered.

Properties

Property 3.1. *If f is differentiable at a, then f is continuous at a.*

Property 3.2. *Differentiability is stronger than continuity: a function may be continuous at a point but fail to be differentiable there.*

Worked Examples

Example 3.1. *Find the derivative of $f(x) = x^2$ from first principles.*

Solution.

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}.$$

Expand the numerator:

$$(x+h)^2 - x^2 = x^2 + 2xh + h^2 - x^2 = 2xh + h^2.$$

So

$$f'(x) = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x.$$

Example 3.2. *Find the derivative of $f(x) = 3x + 1$ using the definition.*

Solution.

$$f'(x) = \lim_{h \to 0} \frac{3(x+h) + 1 - (3x+1)}{h} = \lim_{h \to 0} \frac{3h}{h} = \lim_{h \to 0} 3 = 3.$$

Example 3.3. *Consider $f(x) = |x|$. Is f differentiable at $x = 0$?*

Solution. *Using one-sided reasoning:*

$$\frac{|h| - 0}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0. \end{cases}$$

The left-hand and right-hand limits are different, so $f'(0)$ does not exist. The function is continuous at 0, but not differentiable there.

Exercise Set

A. First-principles practice

1. Use the definition to find the derivative of $f(x) = x^2 + 1$.
2. Use the definition to find the derivative of $f(x) = 5x - 2$.
3. Use the definition to find the derivative of $f(x) = x^3$.

B. Reasoning

4. Explain why differentiability implies continuity.
5. Give an example of a function that is continuous but not differentiable at one point.

C. Mixed return

6. Evaluate $\lim_{x\to 2} \frac{x^2-4}{x-2}$ and explain why this type of limit resembles a derivative calculation.
7. Check whether
$$f(x) = \begin{cases} x^2, & x < 1, \\ 1, & x \geq 1 \end{cases}$$
is continuous at $x = 1$.

Retrieval Prompt

State the derivative definition from memory, and then explain in one sentence why the variable h must approach 0.

Spaced Repetition Block

Planned returns from earlier lessons.

1. Revisit one factor-and-cancel limit from Lesson 1.
2. Revisit one continuity test from Lesson 2.

Integration into new learning.

- Do not separate derivative work from limit work. Ask students to mark the exact step where the limit is doing the real mathematical job.

Lesson 4 Derivative Rules and Applications

Lesson Goals

By the end of this lesson, students should be able to:

- use standard derivative rules correctly,
- compute derivatives of polynomial and rational functions,
- interpret derivatives as rates of change and slopes,
- solve local optimization problems in simple settings.

Definitions

Definition 4.1. *If $f'(a)$ exists, it represents the* ***instantaneous rate of change*** *of f at a and the slope of the tangent line to the graph of f at $x = a$.*

Properties

Property 4.1 (Basic Rules)**.** *For differentiable functions f and g and constant c:*

$$\frac{\mathrm{d}}{\mathrm{d}x}[cf(x)] = cf'(x),$$
$$\frac{\mathrm{d}}{\mathrm{d}x}[f(x) + g(x)] = f'(x) + g'(x),$$
$$\frac{\mathrm{d}}{\mathrm{d}x}[x^n] = nx^{n-1} \quad \text{for } n \in \mathbb{N}.$$

Property 4.2 (Product and Quotient Rules)**.**

$$(fg)' = f'g + fg',$$
$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad \text{if } g \neq 0.$$

Worked Examples

Example 4.1. *Differentiate $f(x) = 4x^3 - 5x + 2$.*

Solution.

$$f'(x) = 12x^2 - 5.$$

Example 4.2. *Differentiate $f(x) = (x^2 + 1)(x^3 - 2)$.*

Solution. *By the product rule,*

$$f'(x) = 2x(x^3 - 2) + (x^2 + 1)(3x^2).$$

Example 4.3. *A particle moves so that its position is $s(t) = t^2 - 4t + 1$. Find its velocity at time t.*

Solution. *Velocity is the derivative of position:*

$$v(t) = s'(t) = 2t - 4.$$

Example 4.4. *Find the critical points of $f(x) = x^2 - 6x + 5$.*

Solution.

$$f'(x) = 2x - 6.$$

Set $f'(x) = 0$:

$$2x - 6 = 0 \implies x = 3.$$

So the critical point occurs at $x = 3$.

Exercise Set

A. Rule practice

1. Differentiate $f(x) = 7x^4 - 3x^2 + 8$.
2. Differentiate $f(x) = x^5 + 2x - 9$.
3. Differentiate $f(x) = (x^2 + 3)(x - 1)$.

B. Applications

4. If $s(t) = 3t^2 + 2t - 1$, find the velocity function.
5. For $f(x) = x^2 - 8x + 7$, find the critical point.

C. Mixed cumulative practice

6. Check continuity of $f(x) = \frac{x^2-1}{x-1}$ at $x = 1$ if $f(1)$ is undefined.
7. Evaluate $\lim_{x\to 0} \frac{(x+1)^2-1}{x}$.
8. Explain how a derivative question may still require old limit knowledge.

Retrieval Prompt

Without notes, write the product rule and then give one original example where you would use it.

Spaced Repetition Block

Lesson-start retrieval set.

- One limit law question,
- one continuity question,
- one derivative-from-definition question.

Delayed return design.

- In the next lesson, students should solve an optimization question and identify which earlier derivative rules they needed.

Lesson 5 Integrals and Accumulation

Lesson Goals

By the end of this lesson, students should be able to:

- interpret an integral as accumulation,
- compute basic antiderivatives,
- connect integration to differentiation in simple cases,
- explain how prior derivative knowledge supports integration.

Definitions

Definition 5.1. *An **antiderivative** of a function f is a function F such that*

$$F'(x) = f(x).$$

Definition 5.2. *The **indefinite integral** of f is written*

$$\int f(x)\,\mathrm{d}x = F(x) + C,$$

where $F'(x) = f(x)$ and C is a constant.

Properties

Property 5.1 (Basic Antiderivative Rules)**.**

$$\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1),$$

$$\int c\,\mathrm{d}x = cx + C,$$

$$\int [f(x) + g(x)]\,\mathrm{d}x = \int f(x)\,\mathrm{d}x + \int g(x)\,\mathrm{d}x.$$

Property 5.2 (Derivative–Integral Connection)**.** *Differentiation and antidifferentiation are inverse processes in the basic introductory sense: if $F'(x) = f(x)$, then*

$$\int f(x)\,\mathrm{d}x = F(x) + C.$$

Worked Examples

Example 5.1. *Compute*

$$\int (3x^2 + 4)\,\mathrm{d}x.$$

Solution.

$$\int (3x^2 + 4)\,\mathrm{d}x = 3\int x^2\,\mathrm{d}x + \int 4\,\mathrm{d}x = 3 \cdot \frac{x^3}{3} + 4x + C = x^3 + 4x + C.$$

Example 5.2. *Find an antiderivative of $f(x) = 2x - 5$.*

Solution.

$$\int (2x - 5)\,\mathrm{d}x = x^2 - 5x + C.$$

Example 5.3. *If the velocity of a particle is $v(t) = 6t + 1$, find a position function.*

Solution. *Since position is an antiderivative of velocity,*

$$s(t) = \int (6t + 1)\,\mathrm{d}t = 3t^2 + t + C.$$

Exercise Set

A. Antiderivative practice

1. Compute $\int (5x^4)\,\mathrm{d}x$.
2. Compute $\int (2x^3 - 7x)\,\mathrm{d}x$.
3. Compute $\int 9\,\mathrm{d}x$.

B. Link to derivatives

4. Differentiate your result from Exercise 2 to check it.
5. Find an antiderivative of $f(x) = x^2 + 2x + 1$.

C. Mixed cumulative practice

6. Differentiate $f(x) = x^3 - 4x$.
7. Find the critical point of $g(x) = x^2 - 2x + 1$.
8. Evaluate $\lim_{x\to 2} \frac{x^2-4}{x-2}$.
9. Explain how old material from limits and derivatives still supports the integral lesson.

Retrieval Prompt

From memory, explain the difference between a derivative and an antiderivative in one or two sentences.

Spaced Repetition Block

Cross-lesson return.

- Ask students to solve one derivative problem before starting integration.
- Ask students to explain why antiderivatives make little sense if derivatives are not already secure.
- End the lesson with one mixed set: limit, continuity, derivative, integral.

Lesson 6 Cumulative Review Lesson

Lesson Goals

By the end of this lesson, students should be able to:

- solve mixed problems across the whole chapter sequence,
- identify which prior concept each problem depends on,
- explain how spaced repetition improved the course structure,
- prepare for assessment with cumulative rather than blocked review.

Cumulative Mixed Set

Exercise 6.1. *Evaluate*

$$\lim_{x\to 3} \frac{x^2-9}{x-3}.$$

Then state which earlier lesson this belongs to.

Exercise 6.2. *Check whether*

$$f(x) = \begin{cases} x+2, & x<1, \\ 4, & x=1, \\ 2x+1, & x>1 \end{cases}$$

is continuous at $x=1$.

Exercise 6.3. *Find the derivative of* $f(x) = x^3 - 2x^2 + 1$.

Exercise 6.4. *Find the critical point of* $f(x) = x^2 - 4x + 6$.

Exercise 6.5. *Compute*

$$\int (4x^3 - 2x)\,\mathrm{d}x.$$

Exercise 6.6. *Explain in a short paragraph how the derivative depends on the idea of limit.*

Meta-Reflection on Spacing

Spaced Repetition Block

Discussion prompts for teaching.

1. Which topic came back most often in this course, and why?
2. Which problems felt harder after delay than they did on first contact?
3. Which returns were most useful: next lesson, one week later, or mixed inside a new topic?

Teacher decision rule.

- If a future topic depends on it, do not let it disappear after the first lesson.

Final Revision Set

Part A: Short retrieval

1. State the definition of continuity.

2. State the derivative definition.
3. State one basic antiderivative rule.

Part B: Mixed application

4. Compute a limit.
5. Test continuity of a piecewise function.
6. Differentiate a polynomial.
7. Find an antiderivative.

Part C: Explanation

8. Why is blocked review less trustworthy than cumulative review?
9. In this course, where did you see spaced repetition most clearly?

Teaching Summary: How the Spaced Repetition Works

Teaching Note

This coursebook models spaced repetition in four concrete ways:

1. **Within-lesson retrieval:** each lesson ends with a short prompt that asks students to restate, recall, or distinguish the key idea.
2. **Next-lesson warm-up:** every new lesson deliberately begins by bringing back older content.
3. **Mixed exercise design:** once a topic has been introduced, it does not stay isolated; it reappears inside later tasks.
4. **Cumulative review lesson:** the final lesson does not treat prior material as finished. It treats it as still active.

Remark. *A calculus course taught with spacing is not just a list of topics in order. It is a sequence in which earlier knowledge remains available long enough to support later understanding.*

Optional Extension

If you want to extend this file, the next natural units are:

- the chain rule,
- definite integrals,
- the Fundamental Theorem of Calculus,
- optimization and related rates,
- exponential and logarithmic functions.

For each added unit, keep the same pattern:

1. definitions,
2. properties,
3. worked examples,
4. exercises,
5. retrieval prompt,
6. explicit spaced review of older material.